

Digiscore India Meeting

ZOOM Jan 9, 2026

Host: Sandeep Bhagwati

Participants: Sameer Dublay (vocals, Flame University Pune), Shawn Mativetsky (Tabla, McGill University), Saulo Olmedo Evans (Tabla), Ninad Puranik (Software, McGill University), Michal Seta (Software, 148 labs), Alberto Acquilino (Software, McGill University, as Guest)

This conversation took place on Zoom on January 9, 2026, almost a year after the research-creation phase of DigiScore India had ended. It brought together several of the musicians and researchers involved in the project to revisit the experiments from a distance: to recall their experiences with the digital-score prototypes, reflect on what had remained significant, and consider their possible implications for Indic art musics. Conceived as a recapitulation rather than a further research experiment, the conversation offered an opportunity to compare perspectives that had emerged during the project with reflections formed after a longer period of distance. The following is an edited transcript of that discussion.

00:00:12

Sandeep: So, thank you for joining. And, being part of this exercise, which is basically an exercise to revisit our work together in the Digiscore India project. Both the first one in which, from the people here, only Samir participated in. And the second one, which everybody else participated in, with the exception of Alberto, of course, who is a guest visitor. And I'll let maybe Alberto, explain a little bit, just in a few seconds, why you are interested in joining us. And, Ninad, if you want to introduce him, that's fine too.

00:01:15

Ninad: Alberto is a colleague from my lab. He'd recently finished his PhD, about Assistive Technologies for music education, basically, and his focus was on trumpet education, but it was very generic in terms of developing applications and tools for evaluating, or creating exercises to learn, etc. And now he's continuing something which is interesting in the sense that he's using generative AI to create some compositions in a way that the students can practice. But the compositions are generated by the LLM, which is nased on input by the teachers. So, in a sense, it's a guided generation. And it's almost real-time. And I think it would be interesting to use such ideas in what we are using. Right now, our generations were just based on, simple rules, and we input the rules, hard-coded rules, but it could be a nice, interesting approach to generate the, like, compositions using LLMs, but then we check them whether they fit our rules, and then only pass them. But anyways, he can talk more about his approaches. I don't know the details.

Sandeep: Alberto, Thank you for joining us, and thank you for your interest in our work. And I would ask you to just be patient with us, because first we go through the past. And discuss things that you obviously were not part of, and we need to do that first, and in the last part of our

meeting, we will come to the future options and future possibilities, future ideas for development of this project. So that would be probably where your input would be most valuable. Is that okay with you?

Alberto: Sure.

00:03:34

Sandeep: As I said in my email. The purpose of this meeting is twofold. One aspect is that I have to write up something about this project for the book that the Digiscopre grant project which finances it is producing. And, I will turn to each of you who has technical knowledge and all that when I'm writing it, to get more ideas about the technical side of it. So we're not discussing the technical side of it today. But what, I'm interested in is the aesthetical aspects, but also the musical memories, the potential for musicians. And, I sent you a number of questions, which I will ask you again. I hope you have had some time to give them some thought yourself. And these will guide a little bit our conversation. I invite you to jump in, not like in a normal Zoom conversation, where we patiently wait 10 minutes until we say something. Let's treat this more like a round table, everybody's mic is open, and we just add our thought whenever we have it. It's a small enough group, it won't cause that much chaos, I hope. And if it causes chaos, I will ask everybody to hold on and retract and start again.

So, I would ask you to open your microphones.

00:05:41.

Sandeep: Let's focus for a moment more on Sean's project, because that is the one where we have the musician who played it and Sean himself here together. So, the first question is not on the list. The first question is, what was the general experience? Were there any moments that stood out for you as being either wonderful, or horrible, or problematic, or whatever - from any side, technical, musical, or compositional.

00:06:51

Shawn: I don't know if this is something we want to be getting into, but in terms of the end result of what the score was, was not what I initially intended. That we had to make a compromise, because of timeline, because of, costs, maybe size of technical team, things like that. We had a bit of a crunch, and we had to compromise and come up with a kind of a plan B. I don't know if that's also part of this discussion.

Sandeep: Sure, absolutely, yeah.

00:07:12

Shawn: I would wish that it could be more than it was, in terms of the response from the computer. I mean, there was response from the computer, but it was less than obvious what the computer's input was in response to what we were doing. Compared to what I was initially thinking. But still, what we had was fun, and it was enjoyable. But I think, playing the score with

with Saulo added a whole other dimension that made it so much more enjoyable: to be able to play with another person, as opposed to just one of us on our own with the computer. And maybe the fact that there was another person helped to make up for the fact that the computer wasn't as responsive as I wanted it to be. Because, obviously, playing with another human being is kind of the most responsive kind of performance we can have, right? With the most, you know, interaction and communication.

00:08:24

Sandeep: I understand that the level of responsivity is, or was, very important to you.

Shawn: Yeah.

Sandeep: Let me just channel this to Saulo immediately. Saulo, how did you feel about this particular aspect, the responsivity of the system?

Saulo: Yeah, I totally agree with what Sean just said. At the beginning, just playing alone. You're floating somewhere in the universe, just playing random rhythms, but then when you play with someone, it actually makes way more sense, because you can adapt to what you hear. And kind of overlay ... try to do something musical about all the information you get from the screen.

But then, I felt also that you have to decide what you want to play on what you're hearing at the moment, so you're using your spontaneity at the same time, playing double speed, half speed, on whatever Sean was playing. So yeah, that was interesting.

00:09:49.

Sandeep: And, Ninad, how did you experience this problem?

Ninad: Talking more mostly on the musical part of it, and leaving aside the technical side: I thought it was interesting that both the players we had were essentially from the same Gharana of Tabla, and they had a similar line of thinking in some aspects, or a very similar repertoire. So I think it was a helpful thing, because a lot of the compositions from Sean were from his Gharana. And the bols were very similar. So it might have been also interesting if it had been a player from a different stylistic background playing together. That was one of my thoughts that day. Another thought was: we did not follow any Tal cycle or a structure that normally is followed. We were kind of shifting from one cycle, or to other cycle, or their playing was mostly linear, it was not cyclic in time, so... But still, it didn't sound too bad. It was still interesting, I felt. It didn't take away a lot of the core values of Indian music. We're still there, in spite of not having this particular thing we consider very important.

00:11:14

Shawn: That was something that I would love to be able to do, is somehow have timekeeping, and so that we can bring it closer to traditional aspects.

Ninad: I understand that as a sort of an expert in the field, but as a pure audience who has never listened to this kind of music, it didn't sound very different or very bad, because most audience

are not able to keep the time as it is when they are listening to a performance. So, to them, it would not sound very bad, I would say. It would be quite similar, in a sense.

00:11:53

Shawn: I guess I'd say it's a more contemporary slash experimental approach.

Ninad: Yeah.

Sameer: It's pretty weird.

Shawn: What we were playing was sourced from tradition, but it definitely wasn't traditional in any way.

Ninad: Right, right.

00:12:08

Sandeep: That kind of lines up with the first written question I had asked you. What kind of artistic imagination could become possible when a digital system creates prompts that are different from what traditional queues ask you to do. You partly answered it already, and maybe Samir can chip in here for the Smruti Ranga piece.

Sameer: Yes.

Sandeep: Again, there, the prompts were not very traditional... And when you started out working with the system, did you experience them as wrong, or what was the kind of attitude that they awoke in you?

Sameer: So, before going into Smruti Ranga, I would like to also share what Charu had to say about Sean's vocabulary. And he said that these exercises, these prompts are useful for doing riyaaaz of the specific rhythmic syllable.

00:13:12.

Sameer: So, Sean, for example, if there is a phrase *Tirakita* And then that phrase is being repeated in, various formats. Then, Charu's idea was that that helps me find my skill as far as the syllable *Tirakita* is concerned

Sandeep: It would have pedagogical implication.

Sameer: Yes, that would mean that if these kind of prompts are generated, to keep the central motif restricted to certain bols. That would generate more possibilities in terms of practicing the, the rhythmic syllables. And that could be, I mean, there could be a million possibilities that you can think of and come up with those prompts. So, that was the feedback that Charu gave to me.

He told me that, when we did it, the line 1 and then prompt of line 2 would be there. And he always invariably said that the line 2 was more interesting than line 1. So, while playing the line 1, he would inevitably think of how he would play line 2. When we discussed the experience, he

said that this is actually very useful for beginners, middle-level, performers, and even the maestros can use this as a tool to practice and make their students practice using this system.

00:15:06

Sameer: Now, coming to Smruti Ranga...

Sandeep: Or maybe, maybe let's just let Sean respond to this...

00:15:14

Shawn: Just very brief, that's all great. I think that this latest version was a response to an earlier version, where we were trying to have Palta of Kaida, which was my original intention, and which in the end, we weren't able to do, but if we are able in the future, I think I would like to go back and re-explore that. And I always said, Sandeep, many times throughout the project: I know this isn't the focus, but I think that this would have great pedagogical applications. With Ninad, we spoke about a kind of pedagogical video game where it could be fun for students to go through different variations that the, the system would prompt. I think there are definitely a lot of pedagogical possibilities!

Sameer: Yes, yes.

Sandeep: for Sean and Saul, and maybe Samir also, and that's of interest to me: It requires a kind of sight-reading, right? This kind of approach. And this is not, of course, not something that is usual practice in Indian music, to have to sight-read something completely new that you haven't seen before, right? So, what did that change in your attitude or behavior, and how do you, Sean, think about it as the creator of this: What would be the role of side-reading in all of this? Or: what is the change that is introduced by this simple technical thing?

00:16:51

Shawn: I think, you know, traditionally, like, notation is not so much part of the pedagogy, but I think, maybe unfortunately, that the students are getting used to writing more, which means they're getting used to reading more. But ultimately, we have to memorize everything anyway. It's not that difficult to read these very short phrases. There's a tradition, you know, where the guru would speak a phrase, and you have to immediately memorize. Here, you're seeing a phrase, and you're essentially immediately memorizing. It's just that instead of it being through sound, it's through visual. It's maybe not so different, because we're not talking about learning Western notation symbols and things like that. It's literally just bols, right? And it's not a huge amount of information, just short snippets So I think it's in a way a similar thing, just a different medium.

00:17:55

Sandeep: Saulo, what did... how did you experience that?

Saulo: Yeah, well, you know, in Western notation, there's so much information, you know: dynamics, the rests etc., and it's very precise. I used to read once upon a time, but I felt that with

the bols, it was easier, because they're short. And there are "sentences" that you repeat all the time in different compositions, so just look at it, and it's already in your hands. You can already play it without needing to analyze what you're actually reading.

00:18:40

Sandeep: So, Sameer, in Smruti Ranga, the whole proposition's different.

Sameer: Yes, absolutely. I mean, for Smruti Ranga, the first, my first reaction, I remember it very vividly, was that this is a real test of my ability to be tuneful throughout the exercise. Because the sequence of notes that were appearing on the screen as a prompt was not the usual one which we are, practicing day in and day out. And which kind of gives us a sense of correctness of the note every time we practice those phrases. But here, the situation was very challenging in terms of the sequence of notes that would appear. And even the, change of octaves was also there. Then, change of octaves along with change of the sharp or flat notes was also there. So, it all really, put me in a position of real concentration and focus on what I am seeing and what I have to produce from my voice. So, this coordination of my visual perception and my tonal perception This, matching of the two was a very great experience for me.

00:20:08

Sameer: Secondly, it also kind of helped me think about my, conventional Raga phrases from a different standpoint. For example, there would be Sa Ga Ma Pa today. So, if this phrase, if you look at it from the Raag idiom, it can be a phrase from the Raag Madhuvanti, for example. I would probably never sing this phrase in Madhuvanti, but if I look at it as a phrase, it is a very much possible phrase for Raga Madhuvanti. So, in a way, it added to my imagination in that particular Raga.

00:21:01.

Sandeep: So you're saying that it proposed things.

Sameer: Yes.

Sandeep: Things that otherwise, would not have happened.

Sameer: It kind of led me into the areas which were very unfamiliar to me. But that was very exciting, because: if you look back to those pitches and check them with the grammatical, matrix of that rag, you can find out: oh yes, this is very much possible in raag Madhuvanti. So, that really helped me re-imagine, in a way, my elaborative techniques in a specific raag that really helped me in being stable at the correctness of note, and be in reimagining the phraseology of the raag.

00:22:05

Sandeep: I find this very interesting, because it speaks to something that is mostly a European proposition: To explore the realm of possibilities of one particular, idiom, or one particular language. And technology, but also just compositional devices, have been used to create precisely this kind of reaching out into other ways of using the language that we have. And so, if that

happened for you in your language. That is a very profound, shift in perception, probably, right? What would you say?

00:22:54

Sameer: Yeah, I mean, that is a shift in perception of the raga possibilities that I have in my tradition. It might be considered as an addition to the tradition, actually. So, that particular phrase, I remember using it in my later performances a couple of times. And that really was a pleasant surprise for many.

Sandeep: Did something like that happen in the rhythm? Well, Shawn's idea was not geared towards that, but: Did something like that happen for you, where you said: oh, this is an interesting way of using a certain combination of bols that I would normally not use, but which are totally within the scope of what I can do traditionally. Or was that not something that you focused on, or something that happened for you?

00:23:57

Shawn: I think the database of bols that I put were not s...I don't know ... far-fetched? Like, you know, I tried to give a good variety, maybe things related to speed changes or things like that, that, you know, maybe we were forced to go from a higher speed to a lower speed that normally we wouldn't do. Normally, we would go low to high. Maybe things... I don't have a specific memory of something where I was like: "Oh my god!" But it's also thatm for myself, you know, I've done a lot of experimental, free improvisation and things. So I'd have to turn that part of my brain off ! So nothing shocked me, but I think if we were able to get the Kaida Palta system working there we would get some variations that would be similar to what Samirji described: "Surprises". Things that are legal, but that maybe we wouldn't have thought of. I definitely think that kind of thing would happen.

00:25:04

Ninad: Yeah, in fact, I remember this happened in one of the trials that we did, Sean, in the CIRMMT room, that you would have said once, like, oh, this is a legal palta, but I would not have played it here. So, something like that, I guess.

Sandeep: So, this touched on the second question already. Could such systems deepen your engagement with the tradition, rather than dilute and make it into fusion music or something similar? Could the systems be helping you to understand the tradition in a more profound or different way? And you have partly answered that question with your responses already.

00:25:49

Sandeep: Now, the third question that I had was related to something that Sean and Saul had mentioned already: namely the fact that you were playing in an ensemble, in your case the minimal ensemble size: Two people. And in Samir's case, it was a bigger ensemble?

Shawn: Yes.

00:26:13

Sandeep: So, the question is, do these digital scores have the potential of leading or creating such ensembles? And is there any way that you imagine this might be expanded or, you know, brought to more interesting ways of creating these ensembles. Could you imagine something like that?

Saulo: Well, something intrinsic that's come to my mind is that it maybe interesting for rehearsal, but ... to do shows for public? I mean, your sightline is on the screen, you're not looking at the people in front of you that you're performing for. So, that would be a bit, maybe, I don't know, strange to just perform and look at a screen, and there's the public there watching, what's going on? They're watching something else. I guess for just rehearsal, it's something different. Like, as we did, just in a room, just us playing and experimenting stuff.

Shawn: But that's very interesting, because it's kind of like a rehearsal tool, an idea generator, improvisation games, a way of getting us to get to know each other musically in a different way. And maybe then we could retain some of that...

I had the same thought as Saulo: we're playing together in an ensemble. But we're not looking at each other at all! When I'm playing with someone, I'm looking at them all the time. But here our eyes are stuck on the screen - also for fear of missing the next information, you know? You look away for a moment and you lose your place because the thing is moving. Which is different from a paper score, where you could look away and look back, because the music is not moving, right? So your eyes can find the spot that you left off. But when it's moving, it's much harder to do that. That's one challenge for me, is to be able to look at a screen *and* communicate with the other musicians. That's definitely something challenging, so... But I like Saul's idea, this kind of idea that it could be something for workshopping and for rehearsal. And it could still have an effect on a final performance without it actually being there.

00:28:42

Sandeep: Hmm. And maybe it could be geared more to producing unusual but "legal" instances of what you are playing - furnishing you with new ideas for the... for the later togetherness.

00:28:59

Sameer: I mean, our experience of doing the first version of Smruti Ranga in a public concert [in Pune in 2015] was very interesting, actually. The audience was also very eager to know what is happening, actually. And if you remember, we did this in the Mazda Hall concert. And it was a proper, well-attended concert. And some people were definitely annoyed: "What the hell are you guys doing?" But that percentage would always be there in any way, I mean... .

00:29:35

Shawn: Did you show the screens in that context?

Sameer: Yes, yes. The screen, the score was projected on a big screen, and we were looking at this screen, we were not even looking at the audience: the audience was behind us.

Shawn: Okay.

Saulo : Oh.

Sameer: Yeah, and it was very interesting because Sandeep had color-coded the phrases. And we had to pick our own color and follow that phrase. And if, while singing that, if the new phrase appears, we have to shift quickly. And, that was quite fun for us. And even, barring the few puritans which are always there to talk to you condescendingly, but the remaining people were quite excited by the prospect of us getting suggestions from the machine which are not pre-composed. Because in India, the common misconception is that Western music is entirely pre-composed and written music. And Indian music is never written. And here is a situation which is trying to bring these two traditions closer through a technological device.

00:30:54

Sameer: So, the only caveat I would like to mention is the technological support that we require for such a smooth kind of presentation. And that could be challenging sometimes in India. Even though we have perhaps the largest penetration of internet in the world, the stability of network, being able to sustain it for the entire time of the concert, is still a challenge for the for the audience, as well as the artists. And organizers definitely would be having sleepless nights in setting up the entire show.

00:31:37

Sameer Dublay: The third point is, can we have a 3-hour concert of this? I don't think so. This could be, maybe 10-20 minutes, half an hour maximum. Then, if you want to go beyond that, then you will need to add Tabla accompaniment in the same prompt.

Sandeep: This is certainly a possible extension of the concept: that we have the melodic part and the rhythmic part, both being prompted from different prompts, or, in some way, a coordinated prompt, so that the machine "knows" both prompts and can somehow coordinate them

00:32:25.820 --> 00:32:32.250

Sandeep: So that, that would be... that's for the future discussion, maybe. There was one point that I wanted to look back to. A question for Samir: Could you imagine Smruti Ranga also being a tool of pedagogy, like Sean and Saulo imagined it for Tabla ? Mainly for riyaaaz, you two had said, riyaaaz for the tabla? But would it also be possible as riyaaaz for vocalists or melody instrumentalists?

00:33:05.

Sameer: Most certainly. This was a great tool for us to do the riyaaaz of the musical note, per se. Being able to sing the notes correctly in the given sequence itself without relying on the phrases we know already – a very important exercise that we need to do. And actually, these prompts actually reminded me of Dr. Ranade's exercise. He had created a card game out of the musical notes. We have 12 notes in the octave. And he would have 12 cards, and there would be a tanpura

in the room at a very low volume, and he would throw a card at us. And you watch the card and immediately sing the note. And this was part of his voice workshop. And that really helped us because that calls for the tonal memory as well as the muscle memory being completely in sync with each other: The moment you see the prompt, you have to sing it immediately. Now this technology has made it possible to give multiple, prompts in one go. I mean, you can have 10 phrases on screen at one point of time. But then that is a very good tool for my riyaaaz, most certainly.

00:34:39

Sandeep: Great. I have another question for Sean and Saulo. Both of you said that sitting there, glued to the screen - you see that as a problem. I, too, have, in my writings, criticized the digital scores for this. I said it was like watching somebody check their email while you're watching them. So it is disconcerting to have a performer's eyes glued to a screen. And my solution for this was making audio scores, so that all information is relayed to you in audio, and that was something that Ninad, in the beginning, actually had wanted to do: He wanted the bowls to be spoken instead of text on a screen, which was technically and time-wise, probably not possible at the moment then. But would you feel that such audio score would work ? Would it be better or worse? Because then you suddenly have things in your ear.

00:35:51

Shawn: Hmm. I mean, I could see the advantage. I could also see it being very distracting.

00:35:58

Sandeep: Hmm.

00:36:00

Shawn: I guess it's what TV hosts do this all the time, I suppose. They're talking, and they have this voice, telling them what to do and what to say. But that would be something to experiment with. I've never had anything other than at most a click track in my ear, or the music of the other musicians, that I am monitoring. So to hear something potentially out of time... Could be, yeah, a little... tricky. So that would get...

00:36:33

Sandeep: The software would need to exactly match your speed.

Shawn: Oh, you think? Yeah, but then would we hear it?

Sameer: It'll be here.

Shawn: If it's so well aligned, I might not even hear it clearly sometimes. I mean, that would be a concern, that you don't hear the nuance, or that you miss something musically important. But I don't know. There's something about visual that, well, because by audio, you wouldn't be able to

have colors, you wouldn't be able to have movement, you wouldn't be able to see the relative size of your playing. You could have volume, of course. You could have pitch.

I'm not sure, yeah. I'd have to experience it once to know what it's like. And just quickly, though, since I'm talking, a point that came up a little while ago with the digital score was that it was a fun experience. Not that it's a game, necessarily, but...that there is a fun to the challenge.

00:37:29

Sandeep: It's true.

Shawn: It's not only discouraging. Like - what is this? It is fun, and we're trying to do it, and there is something to that, too.

00:37:51

Sandeep: But I will ask you one question that you can still answer. I first have to write this text, but I might want to write a grant application where we can pursue this line of work. And is this something that would interest you to pursue further, and to work with me on maybe developing more into something that's workable? Is that something that would interest you in principle?

00:38:24

Shawn: Yeah, I mean, I'm always curious to try things: especially if we can go back to the Plan A around Kaida Palta and how make that happen. What I'm really interested in is that the computer is listening and responding, that it's a two-way communication rather than a one-way communication. I mean, for me, that's interesting. And thank you for, for wanting to continue.

Sandeep: I'm thinking seriously about it, because it's very obvious that we have just made the first small step in all of this. And in order to really evaluate how it might go on, we need at least to take a second step before we can say anything. That might be, you know, a real musical event, or something like it. We still have too little data for anything. Maybe writing this text will give me new ideas about what that might be in the future, but I will ask... all the others, I will ask them now.

00:39:45

Shawn: Sure. It's just that, like, the goal of the first project was specific because it was part of that big grant. And if this is a separate project, then maybe we can explore, pedagogical applications and things that before were just in parentheses could then maybe become a more direct focus. Also: pushing personal creativity and riyaz, for sure.

00:40:17

Sandeep: Yes, that would be a royal road, as they say, to the hearts of musicians: if you can offer them a very practical tool for Riyaz. That's certainly more interesting than saying, we have an entirely new way of making music. I don't know if that is really of interest to so many people

who, who struggle to fulfill the demands of the tradition already: which already is a challenge for everybody. So to say, well, yeah, we can go beyond, is maybe, for some, super exciting, but I think more people would be interested in riyaz and conceptual work.

00:41:29

Sandeep: Michał hasn't been able to intervene so far. How did you experience the work on the Smruti Ranga score? Also: the situation in the performance with Anjana, who's not there today,? And, maybe afterwards, Ninad could comment on that too, because you were in the room and you heard her.

00:41:56

Michał: Yeah, so on my end, this was probably, like, the closest I have ever gotten to Indian music. So, I couldn't really appreciate all these things that were just discussed, the different talas, different traditions, how the the traditional clashes with the digital, and how the digital improves or evolves the traditional things. For me, it was all totally new, right?

I looked at it from the point of view of a very ignorant Westerner who is very deeply rooted in the Western music. I'm kind of familiar with some Western traditions, very familiar with contemporary music very familiar with experimental electronic music, and so, I'm at a completely different end of the spectrum.

00:43:00

Michał: On the other hand I've extensively worked with Sandeep on digital scores in the past, so it was kind of a familiar task for me, making this kind of connection between the music and and the score. And, as was already said, time was short, it was not the ideal way of working, but I was there especially for the technological side of the things. And for me, it was basically a learning experience. So, besides providing the technology, I was learning a lot about the music, the Indian tradition, and the interaction between the musicians, and I loved what I heard. I did see some parallels with my Western experimental music practice.

I definitely felt the tension between, okay, well, this is the tradition, this is something unusual, and from my perspective, I thought that the musicians have learned something, something new. They learned different ways of working with the kind of material that they're less familiar with. But my understanding and appreciation of it stops pretty much there.

00:44:35

Sandeep: Might I ask you, what were the parallels to Europe? Because one of the possible responses to this kind of work is that it will make Indian music just like Western experimental music. And that it, it will become uprooted, and put it into a context where it's just one color amongst many or something. I've heard some comments in that vein. So, just curious, what were the things that you found similar, if you remember them?

00:45:16.

Michal: The improvisation dynamic between people who come from different musical cultures. Right? If I play with a jazz musician, and I've played a little bit of jazz, but that's not my thing. I do what Derek Bailey called non-idiomatic improvisation. So, my language is different from that of a jazz musician, especially if I use an electronic music device that does sound synthesis, and that maybe is less focused on the tonal, in a traditional sense, the Western tonality type of sound, but maybe more on sculpting bigger masses of sound. Our languages are completely different, and sometimes it works, and sometimes it doesn't. It depends on our both sensibilities, and understanding, and willingness to try to make it work. So I sense that a little bit.

Sean teaches also, he has a very pedagogical approach. I know he's a professor, and I felt that in the conversation. So there was this very technical, pedagogical, and, how to say, traditionalist approach: we have to stay within the tradition, and then we can step out here, we can step out there, and things like that. So this was the interesting tension for me that was happening there, which is different from the tradition that I know, where sometimes people are willing to just go all the way out. Completely break what I know, try to do something completely differently, and here it's like: reach out here, and reach out there. I found that interesting. How they are trying things, adding a little bit of color here, a little bit of spice there...

00:47:38

Sandeep: Ninad, you were in this project, of course, as a tech/software person, but you are, of course, a musician yourself. How did you perceive the Smruti Ranga piece from your perspective?

Ninad: I'll talk more about the piece with Matthias and Anjana, the last one. We actually did, I think, 3 performances. The first was more to get them used to this new tool. The second was to refine a few things, and then the third was an actual performance.

In the first one, I didn't tell them what the raag used in the program was. It was, in fact, raag Yaman (or Kalyani in Carnatic music). So they were not told, but it didn't take them a lot of time to guess that the melodies are the phrases from this rag, and they started thinking in that rag. That's what I thought.

Sandeep: Compared with the first version which Sameer had played with in 2015, the decisive change in the programming that Ninad introduced was that he would choose only phrases that are familiar to one particular rag.

00:48:59

Ninad: Essentially, how it was programmed, was that I played the phrases on a MIDI keyboard, which was being recorded, and that was how I seeded the tool, or the progression of the notes. And I think we did that as well for the first test with Samirji. So, once they understood that this is all phrased from this particular raag, they would at times, especially Anjanaji, embellishing it with much more with other gamakas and all that stuff, because she comes from the Carnatic tradition. So, she was not exactly playing the notes as they were written on the screen. There was a lot of embellishment, whereas Mathias, given the plucked instrument with parda-s that he has,

was more true to the notes, and he used more meends while repeating the notes. That's one thing I thought. Also, I felt that he was playing much less in an ensemble: he was generally trying to just play in a supporting role, and only played solo when he got space in between, which was rare. The other thing was: for the last piece, they requested me to program the tool with certain raags, so I asked them what their 2 or 3 favorite raags were. And that's when it got interesting, because they thought there will be phrases from these three raags, but sometimes they thought there are at least one or two more raags in the composition, some of the phrases, they could not relate to either of the three raags, so they had to think in a completely new raag. Which is something Same Samirji also said: that Re he heard was from, like, a transition from something, and it was not Madhuvanti, but it was coming from something else, but it fit in Madhuvanti when you considered it. So, something of that sort was happening with them as well. Anjana-ji told me, it felt as if this raag didn't feel like any of the three raags that we had programmed for it. So similar things came up, I would say.

00:51:26

Sandeep: Did they say anything? Because neither of them is here now. Did they say anything about playing together, how they felt about playing together in this small group ?

Ninad: They didn't elaborate, I would say. They said that it was a good experience in general, but they didn't elaborate too much for me to quote on their behalf. Saulo, did you want to add anything there?

00:51:56

Saulo: If I can add something: I remember something really, really interesting that happened, many times, when playing with Sean and with, two other soloists: the segue between what you hear and what you're seeing. So you're playing something on the screen, and then all your colleagues are switching. And then there's a little lapse of time that you're kind of searching, okay, am I playing this ball on the screen? Oh, he's doing double speed. Oh, so there's a little amount of time that you have to improvise. And you're not really playing what you're seeing, or playing what your mate is seeing. You're kind of knitting around to find something musical, according, you know, to what you read and what you hear. So, this happened, for me. Sometimes, and I felt I was like, oh, I don't know what I'm playing, I'm just...

00:52:59

Ninad: Going with the flow.

Saulo: And I found that was interesting.

Ninad: Also, one more thing to add was that, we should not have really chosen sitar if we had had the option. It would have been better to use a vocalist. Mainly because it is a fretted instrument. On it, it was very difficult to switch raags, and that's what is happening constantly in. this piece. So, we should have chosen a much more flexible instrument, and ideally the voice, which is the most flexible of all.

Sandeep: And who quit on us shortly before we came together.... Yes, that would have been good.

00:53:49

Sandeep: But - harking back to the original version of Smuti Ranga: you didn't have the opportunity to compare the two versions, one where the melodies are from a certain repertoire. And one where, in the earlier versions, the melodies were purposely non-idiomatic, because they were random generated from all the 12 pitches. And, Sameer, but you haven't had the opportunity to compare them, but what do you think about the relative merits of both approaches? What would instinctively appeal to you more for which situation, ultimately?

00:54:30

Ninad: For the purpose of riyaaaz, the first version was more useful. But for the other... I mean, if you want to practice the raga idiom per se, then maybe the second approach would be more helpful.

00:54:58

Sameer: If we are talking about futuristic possibilities, if we could have, ideally, the two options as a selection of the mode, one for riyaaaz mode, open for the Raga mode.

Ninad: And...that would generate the prompts accordingly. If that is possible, technically, it would be wonderful.

00:55:26

Sandeep: Yeah, I mean, we're entering the territory of future thinking right now, so thank you all for these reminiscences and for the evaluations. They're really great. There's so much... I'll look at the transcript again: Interesting material there to suss out for a discussion of what happened. Maybe we can briefly bring in Alberto. You have listened to our discussions now. I don't know how much you followed, because a lot of it was Indian Music stuff, I don't know your background there, but in any case: What was the takeaway for maybe future planning or future ideas about this? What would inspire you?

00:56:23

Alberto: I mean, as a little bit of background, my formation is mainly Western music, so I did, have, like, a master's degree in Italy. And now, with Ninad, I was a bit into technology, and I'm very much into pedagogical applications, so that's the part I was able to follow the most. About Indian music: I am fascinated by Ninad's concerts when he invites me. And, for my current project, I'm really invested, let's say. I just started a postdoc last September, and I am starting now on my first grants, to start having the first little funds to build up credibility. And my idea is more to create a sort of adaptable music book.

In this sense, my postdoc is about music education, technology, and disability studies. So, the main objective is, like, how we can enhance the access to people with disabilities to music

performance, music learning, and accessible musical instruments. I am very much into music learning, so I'm very interested in creating an adaptable musical book: in the sense that, if you want to learn a musical instrument, there are normally, in the Western tradition, many books and methods. But they all assume that you have a normal body, And that you don't have cognitive and the physical disabilities If you do have them it is up to the teacher to adapt. What I want to fight for is to try to destroy this barrier by having a digital musical book that teaches according to different multimodal prompts, with which one can create learning material for literally everyone.

And the good thing about Indian music, there is not any extensive, let's say, written notation. And by this token it already doesn't have all these ableist barriers. And for me, also, as soon as you have this sort of method it would be great, you know, to apply different components, different services that are available for the whole music education community.

00:59:14

Sandeep: Could you, maybe in very brief terms, describe the central working mechanism of your adaptable book. So what does the book do?

Alberto: I'm still in the way of creating. I will have different students working on that the next summer. Up to now, it is an interface in which you enter musical parameters, such as key signature, tonality, musical instruments that you are playing, and the duration of the exercise, and then you have an LLM generating material, generating the exercise: This is output as audio and, as you were discussing before, it is also output as Western music notation. I still have to work on implementing this part, e.g. as a piano roll, or whatever, but as soon as, like, you know, you have an output of an LLM, which is structured. Then it's just a click to change the music notation, and change also the form, the modality. It can be something you can see, it is something you can hear. And in this way, you have, like, a sort of a flexibility also for teachers to decide what is the best learning strategy for their student. For me, it doesn't have to be in the Western music notation and tradition. You should have the potential to extend this to other modalities of other traditions, to different parts. So my idea, the applications I'm working on, are to develop this sort of framework. You send textual prompts, or audio prompts, and you get a musical exercise. And this is a sort of a backend.

Sandeep: The training database, is it local, or is it web-based?

01:01:19

Alberto: I use the person-trained models in the literature, like MIDIGPT, or Music Coco: there are, different labs that they are working on that, like Text2MIDI, or MIDI to MIDI, you have different models that we are relying on. And.

Sandeep: I mean, Indian music notation should be wonderful for LLMs.

Alberto: It's all text.

Sandeep: So, has anyone ever trained an LLM for Indian music notation? Does anyone know this?

Alberto: I would ask also Ninad about that. I am not sure.

Ninad: I know there are people working on that front, and one of them is Nitya Shekarpur, who is working on similar things.

Sandeep: Oh, goodness.

Ninad: She's now doing a PhD at MIT Boston, and she should definitely have much more idea than me about AI. I'm just guessing. There are people working on that at IIT Bombay as well, and multiple places. In Carnatic music, there's a lot of work happening. One of my ex-colleagues from Barcelona is working on that. So, definitely there are things happening, but a lot of it is not yet, I would say, suitable for real-time application context that we are trying to achieve. And it's a bigger challenge, technically, like, to even implement any simple thing in real time. Adaptive, and yeah, that's a very big challenge for things to run in real time. And for us, it was an added challenge to also be distributed, because some of us were not in the same place, some of us were in India. Definitely these things have to be for me, the bigger challenge to achieve: this is not about the models or the research behind it, it is just the execution of any simple model in a reliable way. Because if it's a concert, or if it's a live event, there cannot be a thing going wrong. We don't have to do very amazing things, but we have to do what we do perfectly well. So ideally...

01:04:09

Sandeep: That speaks to something that Samir mentioned. Ideally, any...application would be kind of, almost a consumer-type application that you can have on your phone, right? And which doesn't need any... Technical staff, at all. Right? That would be the end goal for it, obviously.

Ninad: Right.

Sandeep: And because, as Samir explained, of course, there are several logistical hurdles to having such systems being implemented in a large scale.

Alberto: Yeah, I mean, for experimentation, of course, web technologies, they are faster to prototype. It is what I'm trying to do. And another part of what I do: I am trying to validate the model with a disability-co-led design, in the sense that I bring, like, I target a specific disability, and then work with them, create, music learning interfaces, and iteratively test them, and see what is working, what is not working. And, a similar thing, I would say can be done, like, with a community, like, you know, of performers, of teachers. To define which are, like, the musical parameters to feed to the model as prompts. And then, having a significant flexibility, then, for each of them to decide a personalized music path for each student. It can follow, like, the pace or the learning objectives of each student. But also to train the model, it's important to have these musical parameters.

01:05:36

Ninad: There's a sort of labelling needed before. And that would be probably easier in the case of tabla than in the case of raag. But the dataset side is very, still quite incomplete, I would say. For Indian music, especially, the amount of label datasets of any kind are very, very small, if you compare to the Western music. And it's only because of the money invested, I would say. It's because there are so many labs trying to work on building the datasets, and the funding is more. That people are just able to spend 2 years to collect and label data. And that's not happening for Indian music, at least right now.

01:06:25

Sandeep: There's one question, maybe that's a closing question: Is it worthwhile doing that? That's a very fundamental question. I mean, we're all for experimentation and, you know, maybe possible use in pedagogy... but is your gut feeling, in a sense, that this is a path that could lead to something important for Indian music itself? That would make Indian music not better, but maybe richer, or more varied. What would be your gut feeling there?

01:07:42

Saulo: Well, I think in general, in music, as in all directions, to try things is really good. And in all traditions, people have created new ways of playing. Hindustani music before: it was Dhrupad, it had other instruments, and other rules, and then Khayal came, and so there were always new conventions that were created, through time, and maybe at some point, these conventions were considered as a rebel by some musician. I feel in that fact, it's good to try new things. Will it help people to learn Indian classical music better? I don't know at this point. As Sean was mentioning earlier, if ever we include a qaida-structure for tabla, that would make a big difference in the bols' structure, because as of now we're mostly playing just bols on table AND bayan all the time.

01:09:18

Sandeep: Ninad and Samir, any thoughts?

01:09:24

Sameer: I have a strong feeling, Sandeep, that we have here the formulation of skill, information, knowledge, and insight. These are the four levels on which learning takes place in case of Indian music. So, for skill development, for information sharing and even for knowledge, creation. For these three terms used specifically with the reference to Hindustani music, the skill part is how you sit, how you breathe, how you vocalize. Then the notes involved, the raags involved, the bandishes, the compositions and elaboration techniques, these are all pieces of information. The knowledge level is where you start understanding the patterns out of it, or you are able to connect the dots.

Like, for example, if I say Kalyan-ki-prakar, the types of Kalyan, then you know what kind of phrases are meant to be there in that particular tonal framework. If those phrases are there, I can

safely identify it as a Kalyan-ki-prakar, or a type of a Kalyan. These are the three levels that can be achieved through the pedagogical implications that we can see in this kind of device. And to me, it would be very, very interesting tool for music teaching institutions - for the simple reason that in the present scenario, when everywhere, globally, it seems that the fine performing arts are facing an existential crisis. This tool might kind of help in engaging larger population with the musical traditions. And in that sense, it would help get more students. Because if you have a prompt-based, teaching tool for, say, Ragh Yaman. And, maybe there are 10 raags in these prompt data. We can create a certificate course out of that, saying that if you go through all the prompt possibilities and are able to sing and record and send it back to us, we evaluate and give you a certificate in the Raga music - and this without any, institutional infrastructure requirements, only the internet and the user's willingness to engage with this kind of device.

This would open up a lot of, more people accessing the musical tradition from a very different standpoint, actually. So I, I find that it would add to the spread and even the study of that tradition. It would not no longer be restricted to only university-going students, or the students who are primarily interested in learning Hindustani music. Any layman can also think of learning Indian music and then we can also go to our prompts, giving them popular songs, based on that particular raag. Maybe a couple of songs that we can show them as prompts, and give them a chance to do a karaoke singing.

These are all possibilities which I can see which are hidden in this project. That's what I feel: this is definitely a useful exercise, and it would definitely have a business potential as well, if we think in terms of a startup that we can generate from this: riyaz as a service, raag teaching as a service, practice sessions as a service, karaoke singing as service, and this is all technology-based and driven by the AI-generated prompts.

01:14:08

Ninad: I had also one more thought, which is, we are always thinking of this tool as creating new kind of approaches and new kind of inspiring new music, but it has, I think, the potential to actually retain or carry forward the tradition, especially things like uncommon gharana-s. As it is now in Indian music, the concept of gharana is diminishing, and almost everyone listens to and trains to do a variety of gharanas. But the problem with that is very few people can sing a particular gharana without mixing in some of the other's styles. And it's basically restricting yourself to certain phases and not singing certain phrases.

Indian music has always been a very improvisational and changing kind of music. There's little stagnation in the music, but the problem is that certain beautiful aspects of it are changing so much that we are losing out on very important factors or facets of the music. So, one way could be to enforce that the prompts are restricted to a certain style, or restricted to certain, what we call uncommon raags.

01:15:40

Ninad: Then there are certain popular gharana styles, which is Jaipur, Agra, and etc. We still see a lot of singers practicing those, but there are certain others which are losing the number of practitioners. But we could get these experts within their own traditions, and help us train certain kinds of prompts, or verify the prompts generated by LLMs, to determine whether they are from your tradition or not. And then, this tool could be, in some way, used to create modes. You want to train in Jaipur style on this raag, and then the prompts do not go outside of that style.

If your imagination might not be well developed, you might, on your own, mix other styles or other raags. But this tool will help you to stay on track. And this without the presence of a human teacher.

01:16:39

Sandeep: And Sean has this interesting take on it: Could we just analyze the recordings of masters to learn about their individual styles? And that would open up a whole history of styles that are no longer taught, but we still have recordings of them. And it could become a training tool for these styles and, in that manner, these could be revived.

01:17:20

Sameer: I have a slightly different take on this approach. Because none of the master have trained themselves into single style. As Dr. Ranade would always say: "The concept of purity in khayal music is a myth." There's nothing like pure khayal music, or pure Kayal style. You have dominant features of a particular style. That is fine. But sticking to one style and being able to sing it in that style only has never been the case. You take any artist from the 20th century, or the 21st century even. None of the musicians are trained in single style. I mean, they have strong base of a style, and they have inevitably gone on to learn from different maestros and masters of other gharanas. And that has actually enriched their vocabulary in terms of the tonal phraseology. Or the tonal possibilities that they explore. So, I mean, if you take a case like Bhimsen Joshi, even while being trained in Kirana style primarily, had a strong influence of Kesarbai, had a strong influence of, Fayyaz Khazar, and had a strong influence of Jaipur at the end of his life, when he switched entirely to singing in Tintal.

Take any, any other artist, look at Jasraji's example. Look at, Ustad Amir Khan Sahib's example. Dadi Mirashi, kind of a family lineage, who were Sarangi players as a community. But then he went on to develop the Indore gharana, as it is called, but which is basically a combination of Gwalior and Kirana with the Agra vocalization. So, the very idea of having a stylistic revival is a non-starter, according to...

01:19:46

Sandeep: But, you know, I was thinking precisely of Dr. Ranade's take on this, when I said that an individual master has this kind of weird mix – and by training a system on these recordings you could evade precisely this kind of narrow definition of what it is that they're doing according to gharana, and to understand what they were really doing by having a system emulate this again.

01:20:26

Ninad: But this is just a very far future, but now still, it's very interesting to think about that. In the sense, like, for any normal, like, person like me who is just learning music, I would say, just beginning to perform at concerts. I don't have this rich knowledge basis that Sameerji is talking about. Like, when I hear a phrase, it's very difficult for me to predict, is it from this gharana, or this style, or that style?

Sameer: Why do you need that?

01:20:57

Ninad: It is because, let's say, if I listen to some masters of a particular style, there's a particular flow to the music. One phrase follows the next in a certain style or pattern. And that is often missing in some of the performers I see, or even in myself. I would say that there's no linear thought. Now, this could be something you take from your guru, which is fine - but then it is only one thought that comes to you. But can you, like, specifically train on certain phrases for some time using a tool? And thus start thinking in that fashion? That's something interesting, as an experiment - and, as Sandeep said, you could also train a tool to generate or think like a specific person, or a specific master - and you get all the prompts in the style of that person...

01:21:54.

Sandeep: I was thinking of a new phenomenon that is greatly discussed in Western classical music: that the new generation of musicians who are growing up with YouTube and Spotify with their deep archives, have a wide variety of stylistic approaches towards the pieces that were always played in the repertoire. But they developed their taste not from listening to a dominant school: rather, they had all the resources of YouTube and all the recordings of masters both from 100 years ago, and from now. And so they concoct their interpretation as much from the tradition that they're taught in and from the listening experiences that they got by listening to this great variety of music. And it gives us interpretations that are really astonishingly new and interesting. Because they mix and match all these different aspects, they may take two seconds from this interpretation. But they combine it with two other seconds from another interpretation, and that gives us an entirely new insight into the music. And so, I imagine that something like this must be happening in the Indian context, too. That a generation of technologically trained musicians has a very different approach to what a gharana is. What does it mean to have a lineage?

01:23:32

Sameer: The very idea of gharana now is perceived in the Indian discourse as a basic discipline required to learn Khayal, which is largely dominated by Gwalior right now, when it comes to learning. Because the pedagogical insights were developed mainly in Gwalior gharana through the Gandharva Mahavidyalaya tradition, where institutionalized, Hindustani music training was started. And they had to kind of create learning material, they had to create various basic

phraseologies of different raags, and had to notate the compositions. All that is done out of compulsion to teach music in a specific manner.

The rest of the gharana-s, however, have mainly stayed in their discipline through familial teaching. And the lineage was basically biological family. Later on, it has branched out to other, I mean, non-biological followers of the gharana: adoptive children.

A third, very important part about the gharana tradition is that all the gharana-s became famous and started getting recognized as separate stylistic variations only when the masters came to Mumbai or Pune. That's a historical, detail which is very, very interesting, actually. This is in the early 20th century. And if you, if you look at the writings of Bhatkande-ji, for example. Bhatkande-ji diaries were published, and Dr. Ranade was editing the diaries in which Bhatkande-ji mentions that Alladiya Khan claimed that he would sing 21 kinds of kannada-raag in Jaipur gharana. And then he writes: "But he has never been able to show me even 5." That kind of environment was there in the gharana idiom.

Also: a gharana, per se, is built on a discipline of voice training. And for that, the precise phrases of a raag is not what matters. What matters is the vocal technique. So the skill level is what the granas primarily dominate their presence in. I learned to sing in a specific manner, because I'm trained in that particular gharana. So this entire discourse on gharana came back to my mind because I had worked extensively on the gharana tradition while doing my doctoral research a few years ago. I have spoken to all the masters, I have had fortune to talk to the greats, like, Sharaf Hussain Khan from the Agra gharana, Ashwini Tai and Shruti Sadolikar from Jaipur. And Bhimsen-ji himself from Kirana gharana. So, all that has made my belief even more firm, that this is for a very basic learning with a master for at least 5 to 10 years, that you get the foundation, laid properly. When that is done, the remaining part of the development of the artist is definitely from multiple sources. It cannot be single source.

01:27:45

Ninad: I understand your point on that, and I also raised another thing which was about, training prompts for Anava ragas, which you then don't get to hear much, or there are many few people who can teach those rags, and that could be a, maybe, interesting application in that case.

Sameer: Yes, definitely. I mean, anava raag-s, unsung raag-s, as they are called many times, achop rag, as they are called. Now, why does a particular rag become anava ? There is a reasoning: The aesthetic appeal and aesthetic possibilities are very limited in acharchit raags.

Like, for example, if you take, Hindol Bahar. If you look at the repertoire of anava raags, if Hindol Bahar is to be sung, you need to know Hindol very well, as well as Bahar very well independently. And then you can get prompts to... To combine the two.

Or Lalit Hindol, for that matter. Very rare rag. Only, I mean, I have heard Abhisheki-buva singing that sometime. And then, he taught us, the raag and bandishan. But, inevitably: Very limited repertoire in terms of the possibilities that you can create through alaap and other musical material that one can create: That's when they start to become rarely sung raags. Because for a

performer, what matters is if I'm able to get more and more prompts in, more and more training, and that's the reason why a performer is trained. If you look at all the anecdotal history that is available. Every great maestro always said that I was taught Bahar for 6 months, for example, for years, I was learning Yaman. That's what Kesarbai Kerkar, says, with reference to, training method. teaching Yaman for 2 years has something that we need to understand. Why so much of time is devoted to a raag which is so widely known and widely sung: Because it has a great residue in the social and cultural memory of the Indian mind. And that's why the more and more possibilities are generated.

01:30:48

Sandeep: Sorry to interrupt, but listening to you, I think this is exactly, as Nina had said: maybe there's a model that could then create new bandishes in unknown raags, and new options of what we can do with this.

Ninad: Yes.

01:31:03

Sandeep: I was also thinking of the Melakarta system, where you have all these possibilities, and hardly anyone uses them, right?

Sameer: Yes.

Sandeep: And it might be something where the software might propose things that you could do with these unexplored potentials.

Ninad: Right.

Sandeep: And find a new way of using them that has not been proposed by anyone, particularly for the reasons that you say. Cultural resonance. Why do we use a keyboard for computers that is totally outdated and doesn't really fit ergonomically? Because everybody has been using it for 100 years, and we cannot relearn that. And so, it may be similar with the raags, right? Everybody has learned so many of those popular raags that others remain unsung, because no one ever bothered to explore their options.

01:31:59

Ninad: And the possibilities are limited because of the chalan of the raag.

Sandeep: And this challenge could be modified. Or can even propose another chalan for the raag, something that is also possible in its way of movement...

01:32:22

Ninad: Maintaining the chalan becomes the core of the raga identity.

Sameer: The raga-ness of that tonal framework entirely depends on the maintenance of the chalan of the raag in Hindustani tradition. As against in Karnatic music, where this idea of chalan is completely absent.

01:32:41

Sandeep: So maybe these systems are more adaptable for the carnatic music.

Sameer: Karnatic music: this would be very, very evocative in terms of the possibilities.

01:32:56

Ninad: Well, the intense discussion is very useful to me.

Sandeep: Yes, wonderful. Thank you so much, thank you.